

# GPTs in Mafia-like Game Simulation

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Figure 1: Munyeong Kim (b. 2000), ‘AI, Human, and Boardgame (2024)’, ChatGPT prompting.

## ABSTRACT

In this research, we explore the potential of Generative AI models, focusing on their application in role-playing simulations through Spyfall, a renowned mafia-style game. By leveraging GPT-4’s advanced capabilities, the study aimed to showcase the model’s potential in understanding, decision-making, and interaction across scenarios. Comparative analyses between GPT-4 and its predecessor, GPT-3.5-turbo, demonstrated GPT-4’s enhanced adaptability to the environment, with significant improvements in posing questions and forming responses. However, challenges such as the model’s limitations in judging and suspecting actions of other players emerged. Reflections on AI’s future capability and directions

were also discussed. The findings suggest that although GPT-4 exhibits promising advancements over earlier models, there is potential for further development through expanding data and training techniques. The findings also underscore the importance of maintaining an inclusive and unbiased approach throughout this process, suggesting immense potential for Generative AI and its application.

## CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → **Artificial intelligence**.

## KEYWORDS

Generative AI, Role-playing Simulations, Spyfall, GPT-4, GPT-3.5-turbo, Game Strategy, Decision-making, Natural Language Processing, AI in Gaming, Limitations of GPT

## ACM Reference Format:

Munyeong Kim. 2024. GPTs in Mafia-like Game Simulation. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*

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CHI EA '24, May 11–16, 2024, Honolulu, HI, USA  
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ACM ISBN 979-8-4007-0331-7/24/05  
<https://doi.org/10.1145/3613905.3647958>

(CHI EA '24), May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3613905.3647958>

## INTRODUCTION

Current generative AI models, such as GPT-4[12], demonstrate potential for human-like decision-making and simulations[3, 11], rapidly advancing with increased data[9], though dependent on significant financial investment for their development[5]. Excelling in language tasks, these models often surpass human efficiency, opening avenues for sophisticated simulations and non-playable characters in games, providing cost-effective and secure alternatives to human participation. Recently, several studies explore generative AI's capabilities in simulation, indicating notable improvements in human-level reasoning and context understanding of AI[13, 17].

Our study, which utilized GPT-4 to play Spyfall[16]—a social deduction game[7] similar to Mafia or Werewolf game[14]—showcases the model's proficiency in handling complex, language-based situations. This research enhances our comprehension of generative AI's capabilities, particularly in simulating role-playing and social deduction scenarios, and underscores its applicability in both academic research and practical applications.

## GPT-4 AND SPYFALL

Our study used Spyfall, a social deduction game akin to Mafia or Werewolf, to evaluate the linguistic, logical, and social abilities of GPT models.

Unlike games such as Chess or Go, which have a finite number of choices per turn, Spyfall heavily relies on the infinite possibilities of natural language conversation. Additionally, it requires players, whether human or AI, to deduce each other's roles and the game's location, necessitating a blend of social and psychological elements like bluffing, along with significant linguistic and logical skills. Consequently, Spyfall offers a unique approach to assessing AI, focusing on its capabilities in natural language interaction, complex decision-making, and psychological strategies. This contrasts with traditional AI research games that emphasize tactical abilities[4, 10, 15].

## Game Rules, Basic Strategies, and Action Request form of Spyfall

Spyfall, a game similar to Mafia, involves players trying to identify a hidden spy while the spy aims to discover the game's location. All players, except the spy, know this location. The game progresses through rounds of questioning, where the spy guesses the location based on the others' answers, while the other players try to identify the spy. In our study, we used a modified rule for the game, changing the time limit to a turn limit. The GPT models in our study responded based solely on the prompts of the game's rules, basic strategies, and game progress, without the use of additional algorithms or fine-tuning. For the complete rules and basic strategies script used in the paper, please refer to this GitHub repository ([github.com/MunyeongKim/Gen-AI-in-Mafia-like-Game](https://github.com/MunyeongKim/Gen-AI-in-Mafia-like-Game)).

## THE EXPERIMENTS

To demonstrate the competence of generative AI and the potential of generative AI-based simulations, we conducted two main experiments.

- The first experiment sought to assess the extent to which performance has been enhanced through GPT-4 compared to GPT-3.5-turbo. In this test, we explored the test comprised of various topographies of the game and made comparative analyses of the performance among them.
- The second experiment involved playing several games exclusively with GPT-4. We engaged in 8 games with five virtual GPT-4 players, recorded logs from each game, and discussed what the results of these games might imply.

For the complete experiment results and game logs, please refer to the aforementioned GitHub repository.

## EXPERIMENT 1: COMPARISON BETWEEN GPT-3.5-TURBO AND GPT-4

To assess GPT-4's improvements over GPT-3.5-turbo in terms of adhering to format, game context comprehension, and human-like responses, our experiment starts by analyzing the first question of the first turn. This approach, focusing on the game's most consistent and least variable segment, allows for a precise evaluation of each model with minimal external influence.

## Methods and Scripts

We compare the responses of 30 first-turn questions of GPT-3.5-turbo and GPT-4 for each of the 30 locations stated in the "Rules" script. The Action Request Script to ask both models is the same, only changing the keyword of the locations. The Rule and Basic Strategy are the same as the scripts mentioned above, and the response was obtained by combining the three scripts into a request as shown in the figure. For a more accurate comparison, all requests were fixed for player 1, and player 1 was assumed not to be a spy. Thus, the script presented to each model is as follows:

```
"""
(rule script)
(basic strategies script)

You are player 1.
You're not a spy;
the location is '{          }'.
```

```
Now it's your turn to ask a question.
Choose 1-5 players other than yourself, and write a
question.
Please write your response in the form
(n_player,"question")
"""
```

That is to say, "the appropriate question" is the question ensuring those qualities: the model (player 1) should prove that the player itself is not a spy showing that it knows where the location is while keeping the spy unaware of the location. At the same time, the model must follow the format of (n, "question"). If the model

doesn't adhere to the format, it will require significant effort for us to correct its wrong responses into the right ones.

#### [Appropriate Responses]

- (2, "Do you need a ticket to be here?")
- (4, 'How do you usually pass the time during your stay here?')
- (2, 'What type of food do you usually eat here?')

#### [Inappropriate Responses]

- To player 2: "What is your favorite thing about this place? // cf. Not a tuple format.
- Sorry, as an AI language model, I am not capable of playing a game that requires multiple players. Can I assist you with anything else? // cf. Not provide a question for any players.
- (2, ``What is your favorite war movie?``) // cf. The question is not related to the game progress.

## Results

In our experiments comparing GPT-3.5-turbo and GPT-4, as shown in Table 1, we observed a higher error rate and a broader range of response issues with GPT-3.5-turbo, whereas there were no such mistakes with GPT-4. The errors with GPT-3.5-turbo included out-of-context questions, unformatted responses, and direct references to location keywords, all of which disrupted the flow of the game.

Due to these high error rates and the complexities in automatic corrections, we concluded that GPT-4 is more suitable for future experiments and decided to use it exclusively in subsequent tests, moving away from GPT-3.5-turbo despite its lower cost.

## EXPERIMENT 2: RESULTS FROM 8 GAMES BY GPT-4 AND THEIR LOGS

We conducted 8 games using GPT-4, following our discussed rules and capturing logs for each session. All game responses were from GPT-4, and the automation codes were written in Python. Our experiments can be easily reproduced using the rules and scripts on our GitHub.

### Method

We selected locations for a total of eight games. Each game commenced with Player 1's turn. The model (GPT-4) responds to each request independently, without retaining memory from other players or across games.

### Results

Our study found that the dialogue from GPT-4 players was fluent and organic throughout the games. Without specific training, GPT-4 was able to ask varied and relevant questions, suggesting its strong ability to understand and adapt to the game's context.

### The Play of the Game

Particularly impressive was the spy's (GPT-4) performance in the "Crusader Army" scenario.

```
[gamecode: 'ena7009']
```

```
[spy: 5, location: 'crusader army', spywin: True]
```

(...) In the 4th question, player 4 asked player 5 (spy) the following.

"What kind of leadership structure is present in this location?"

To this question, player 5 (spy) responded with the following.

"The leadership structure is hierarchical, with a clear chain of command and different ranks among the personnel."

At the end of this question, no one accused anyone else of being a spy.

The spy did not reveal their identity.

In the 5th question, player 5 (spy) asked player 1 the following.

"What kind of communication equipment is used at this location?"

To this question, player 1 responded with the following.

"Various forms of communication equipment are used here, such as messenger birds, signal flags, and occasionally messages

being transported by horseback."

At the end of this question, no one accused anyone else of being a spy.

(...)

Player 5 reveals that they are the spy, and they guessed the right location (Crusader Army). The spy wins.

Early in the game, the spy had deduced that the potential location was either a "military base" or "Crusader army" based on the other players' responses. In a critical move, when asked about command structure, the spy's response—"hierarchical with a clear division of command roles"—fit both options, avoiding suspicion. Then, by asking about the communication system, the spy cleverly differentiated the two options. Player 1's mention of "messenger birds" inadvertently revealed the correct answer, despite further probing, putting the spy significantly ahead in the game.

### Limitations

However, non-spy players often focused too much on keywords and hesitated to accuse the spy, even when it made suspicious remarks.

Particularly impressive 'worst' was seen in the first game where they didn't use the vote function, even when the spy made an incorrect answer far from the keyword 'airplane', like "enjoying various activities at an 'airplane'."

**Table 1: Error counts by GPT-3.5-turbo and GPT-4, respectively.**

| Description                                              | GPT-3.5-turbo | GPT-4 |
|----------------------------------------------------------|---------------|-------|
| Total Response                                           | 68            | 30    |
| Error Occurred                                           | 68            | 0     |
| Not provide a question for any players                   | 38            | 0     |
| Contextual Errors                                        | 12            | 0     |
| Not game-related questions                               | 11            | 0     |
| The location keyword is stated directly                  | 6             | 0     |
| Format Errors                                            | 68            | 0     |
| Include message rejecting response                       | 62            | 0     |
| Not contain the tuple of (n, “ ”) format                 | 54            | 0     |
| Interaction (e.g. Question and answer) within a response | 2             | 0     |
| Multiple questions in 1 response                         | 3             | 0     |
| Just describing game rules                               | 1             | 0     |

[gamecode: 'ena5001']

[spy: 4, location: 'airplane', spywin: False]

(...) In the 3th question, player 3 asked player 4 the following.

"How do you usually pass the time during your stay here?"

To this question, player 4 responded with the following.

"I usually enjoy watching or participating in various activities and events, maybe grab a refreshment and chat with friends to pass the time."

At the end of this question, no one accused anyone else of being a spy. (...)

This lack of suspicion, possibly influenced by the model's neutrality with the stance 'Well, that can happen,' indicated a shortfall in the non-spy players' strategy. This contributed to a higher win rate for the spy, observed as seven to one in favor of the spy in our experiment.

## To Overcome the Limitations

Yet, addressing GPT models' limitations in non-verbal cues and strategic thinking is achievable with proper training and data. Implementing 'human-like' plays and moves and incorporating non-verbal elements, would improve AI strategy and enhance immersion in simulations. With continued data training and interdisciplinary collaboration, AI can evolve to perform more human-like roles in various role-playing scenarios such as games and education.

## DISCUSSION

### Capabilities of Generative AI and Future Directions

The rapid advancement of Generative AI is creating vast opportunities across diverse fields. Its user-friendly design, facilitating the easy interpretation of outputs in natural language, enhances its accessibility and aligns with Explainable AI principles[1, 8]. This accessibility broadens the appeal of GPT models beyond traditional

tech sectors and reduces the need for specialized programming skills.

Furthermore, the ability to produce responses that closely resemble human interaction is highly valuable in contexts where the quality of engagement is more important than achieving the perfect outcome. AI's capacity to mimic human thought processes and stimulate critical thinking can lead to improved results in scenarios that demand interactive communication.

Looking ahead, with adequate resources, developing versatile simulation AI is feasible. The growing interest in generative AI, fueled by innovations like ChatGPT, suggests a future of continued advancements and practical applications, reminiscent of the enthusiasm that once propelled space exploration.

## GPT and General Purpose Technologies (GPTs): Potential, Challenges, and the Role of 'Big Entities'

The phenomenon that a model's performance is determined by 'the more and better data and training' raises questions about how to collect, train, and distribute profits from such data.

Personally, I advocate for Generative AI to adopt a broad, diverse approach rather than a narrow, uniform one, aligning with OpenAI's view on General Purpose Technologies (GPTs)[2] and their GPT models[6]. A transparent, collaborative development process involving open-source AI and academic sharing is essential for fair distribution across various positions. This inclusive approach not only conserves resources by avoiding redundant AI investments but also fosters a more sustainable AI industry. While major entities like governments and tech giants play a crucial role in this development, this strategy benefits everyone within the AI ecosystem and enables larger entities to expand their influence and remain at the forefront of AI advancements.

## CONCLUSION

Our study on the GPT series of Generative AI highlights significant advancements in AI models for social and role-playing scenarios. Through the game Spyfall, we compared the context understanding, strategic play, and psychological abilities of GPT-4 and GPT-3.5-turbo, finding that GPT-4 outperforms its predecessor. However,

GPT-4 still faces limitations, such as an inability to interpret non-verbal cues and a tendency towards a neutral, unsuspecting approach.

In conclusion, our research underscores the importance of careful and deliberate progress in AI development, with an emphasis on wide applicability and addressing interdisciplinary challenges. The path forward should focus on endowing AI with more authentic human-like qualities, thus expanding the scope of AI's capabilities. The trajectory of Generative AI, paralleling the evolution of video games and underscored by the rise of ChatGPT, suggests immense potential for the future.

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